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ARGENTINA 2026

LATIN AMERICA HIGH PERFORMANCE COMPUTING CONFERENCE

September 21-25

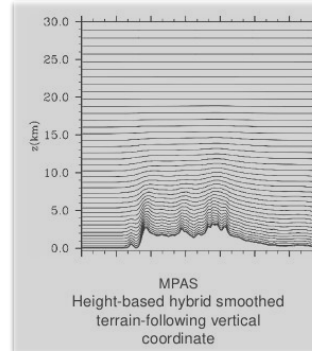
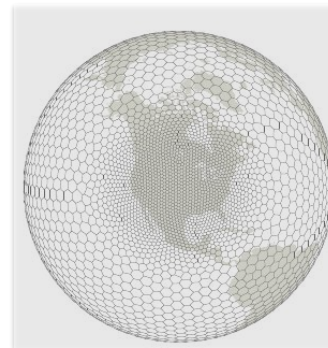
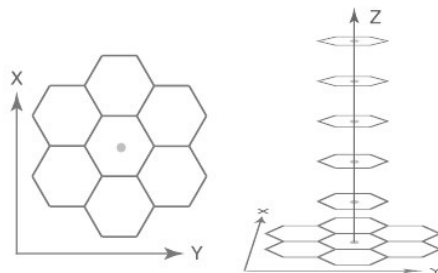
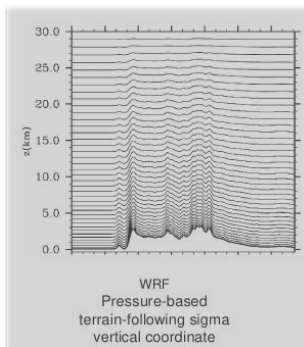
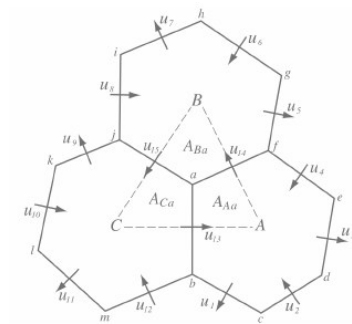
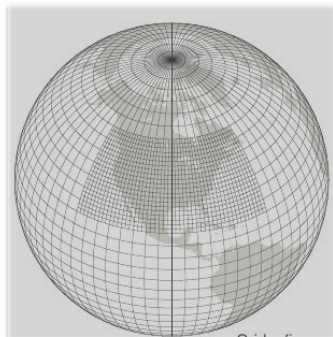
Operationalizing MPAS-Atmosphere at IDEAM: Regional Mesh Design, Workflow Engineering, and Computational Resource Characterization

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IDEAM, Colombia

Presentation outline

- IDEAM Forecasting baseline and implementation goal
- Regional mesh design
- MPAS-A build and forecast workflow
- Computational characterization
- Operational lessons and adjustments
- Forecast product generation

WRF to MPAS-A



Numerical weather prediction at IDEAM

- Current WRF based forecast system uses GFS forecast data.
- 10 km resolution over Colombia, 5 km in the central region and 1.6 km over Bogotá.
- The MPAS-A implementation targets a seven-day regional forecast with hourly output ($<10\text{km}$).
- Initial development used a single node with 64 CPU cores.

MPAS-A Implementation

- Initial tests: manual
- Automation: three Bash scripts
 - Library downloads: version selection
 - Library builds: compiler flags, build variables, manifest
 - MPAS-A compilation: stack, build ID, PIO/SMIOL (v8.1), debugging options, processor count, manifest
- **SMIOL** library option: successful execution
- **PIO** library : successful compilation; runs completed, froze, or stopped at different time steps
- Other stacks: successful compilation; random runtime failures
- Failed runs: recompilation with debugging enabled
- GFS preprocessing: WRF compilation
- Library stacks: activation by sourcing a Bash script
- Stacks Manifest:

```
Stack MPAS GNU-MPICH
Fecha de instalación : Tue May 26 02:02:00 PM -05 2026
Instalador           : 01_install_mpas_stack_gnu_mpich402_simple.sh
SOURCE_DIR           : /modelos/Fuentes_soft/Stk_mpas_FRM/
INSTALL_DIR          : /modelos/mpas_libs_gnu_stack_mpas_FRM
MPICH_VERSION        : 4.0.2
ZLIB_VERSION         : 1.2.11
HDF5_VERSION         : 1.13.2
NETCDF_VERSION       : 1.12.3
NETCDF_C_VERSION     : 4.9.0
NETCDF_FORTRAN       : 4.6.0
BUILD_PIO            : SI
PIO_TAG              : pio2_5_9
```

```
Compilación MPAS-Atmosphere
Fecha                : Wed May 27 10:06:17 PM -05 2026
MPAS_TAG             : v8.1.0
MPAS_BUILD_DIR       : /modelos/MPAS/builds/MPAS-Model-v8.1.0_gnu_stack_mpas_FRM_mpas8.1_makeModif_smiol_op
t
STACK_NAME           : gnu_stack_mpas_FRM_mpas8.1_makeModif
STACK_DIR            : /modelos/mpas_libs_gnu_stack_mpas_FRM
I/O layer            : smiol
DEBUG_BUILD          : NO
USE_PIO2             : NO
NPROC                : 32
NETCDF               : /modelos/mpas_libs_gnu_stack_mpas_FRM
PNETCDF              : /modelos/mpas_libs_gnu_stack_mpas_FRM
PIO                  : NO DEFINIDO
MAKE_OPTS            : gfortran
```

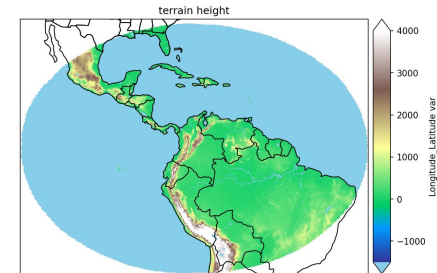
Running MPAS-A focus in Colombia

- Testing MPAS working with 480 Km Quasi-uniform mesh (2562 columns)
- Next: 60-15 Km Mesh x3 : 20-5km (**535,554 columns**), 46 procs, 30 s time steps

(Rule of thumb~> 6 x finest resolution)

⇒ 120 steps/hour. test Result: Average time step integration ~13 s

- Rough estimate:
- ⇒ 120 steps/frst hour x 13s pc time = 1560 s / fosrecast hour
- ⇒ ~ 26 pc mins / Frst hour.
- ⇒ 10h + 24 mins / frst day. ⇒ **Not Viable.**

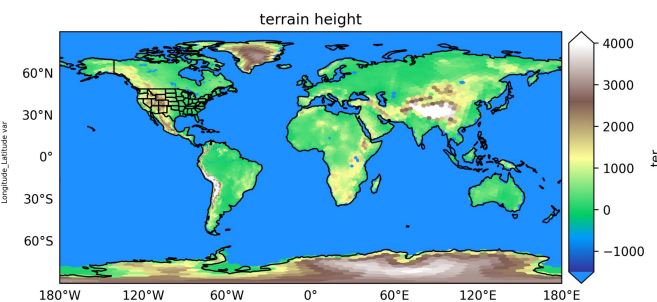
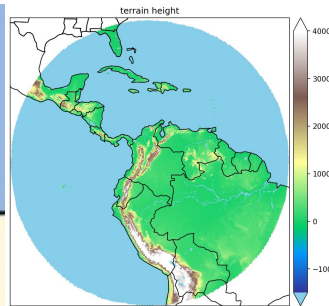
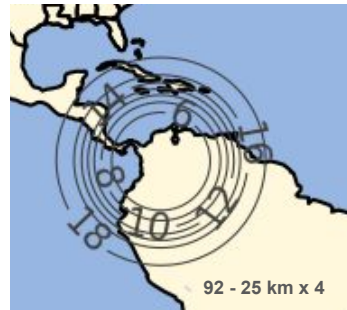
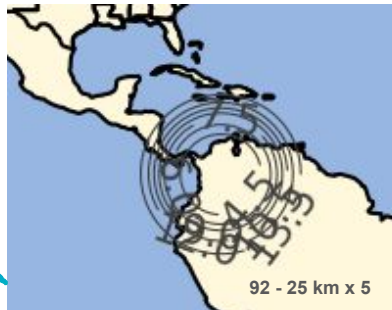


Running MPAS-A focus in Colombia

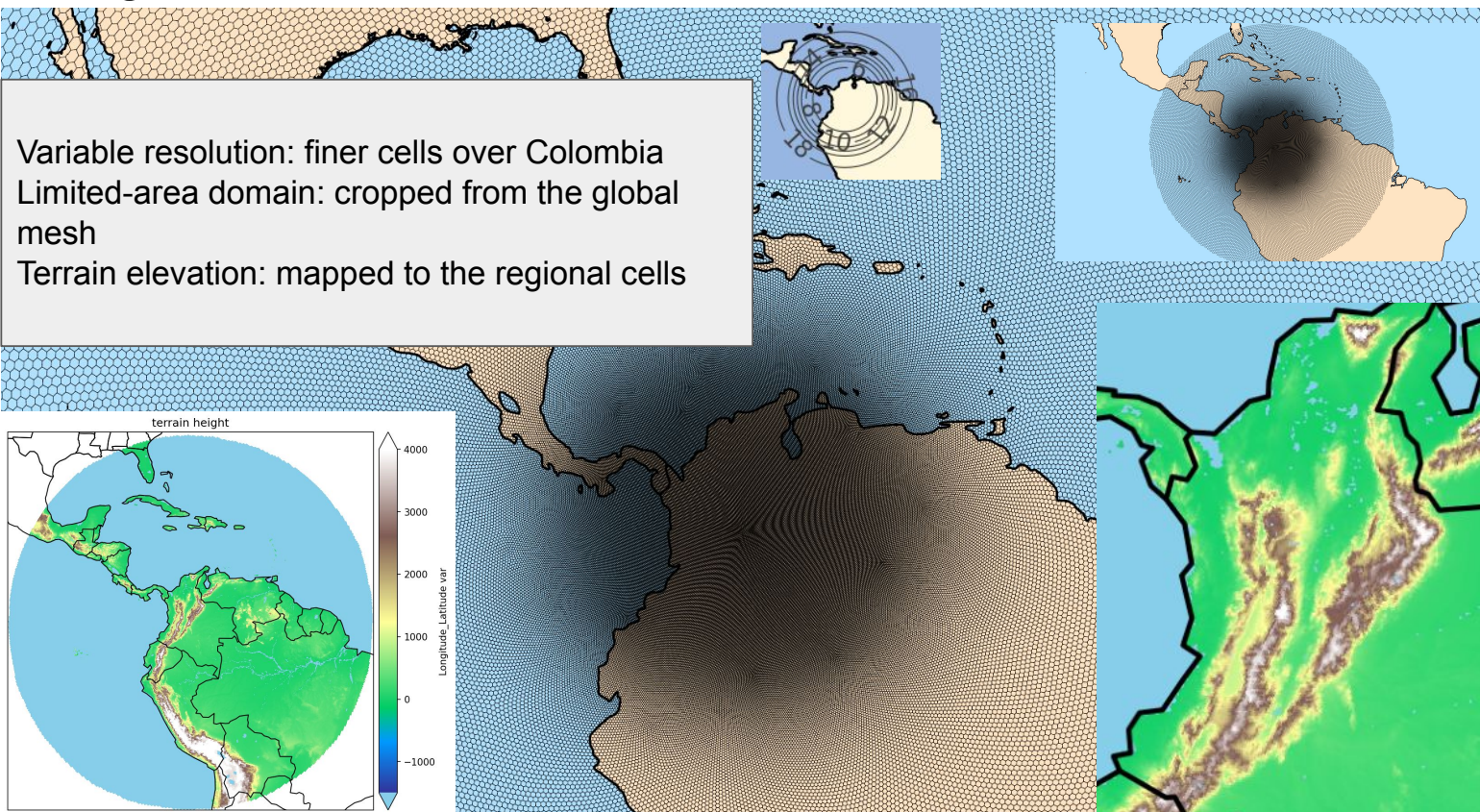
Next: 92 - 25 km mesh (**163.482 columns**), **lightest variable-resolution mesh available**. 92 - 25 x 4 : ~ 23 - 6 km, 30 s.

test Result: Average time step integration ~3.6 s

- Rough estimate: .
- $\Rightarrow \sim 2\text{h} + 50 \text{ mins} / \text{frcst day.} \Rightarrow \sim 20\text{h} + 10 \text{ min} / 7 \text{ days.}$



Regional MPAS-A mesh and static fields



MPAS-A forecast workflow: design principles

Design system characteristics:

- **Modular:** independent stages for testing and diagnosis
- **Configurable:** operational settings separated from execution code
- **Maintainable:** simpler updates and handover to other IDEAM staff
- **Portable:** Bash and standard Linux tools; transferable to other machines as the system grows
- **Traceable:** inputs, logs and outputs organized by forecast cycle

MPAS-A workflow and Weather Forecast system

Routine	Function
config_sis.bash	Declares and exports paths, forecast parameters, MPI resources, and control options.
funciones.bash	Defines functions for validation, date and cycle selection, directory creation, blocking or pausing execution if another cycle is in progress, and preparation of the execution directory.
preparar_gfs.bash	Downloads GFS data, links the GRIB2 files, and runs ungrib to produce intermediate FILE:* files.
generar_archivos.bash	Writes the configuration files for the MPAS-A model and the WPS (WRF) module, namely: namelist.wps, namelist.init_atmosphere, streams.init_atmosphere, namelist.atmosphere, and streams.atmosphere.
ejecutar_mpas.bash	Orchestrates the complete workflow, runs the model for each phase of the process, and organizes the outputs.

```

/modelos/W_frcst_MPAS/
├── Scripts/
│   ├── config_sis.bash
│   ├── funciones.bash
│   ├── preparar_gfs.bash
│   ├── generar_archivos.bash
│   ├── ejecutar_mpas.bash
│   └── limpiar_sistema.bash (en desarrollo)
├── Envs/
│   ├── load_mpas_gnu.bash
│   └── w_env_wrf.bash
├── FCST_OP/
│   ├── Inputs/
│   │   ├── Static/
│   │   ├── GFS/{00,12}/YYYYMMDD/
│   │   └── UNGRIBED/{00,12}/YYYYMMDD/
│   ├── Ejec/{00,12}/YYYYMMDD/
│   │   └── Logs/
│   ├── Outputs/{00,12}/YYYYMMDD/
│   │   └── Productos/
│   └── Logs/
└── 00 -> logs de la ejecución 00 más reciente
    12 -> logs de la ejecución 12 más reciente

```

MPAS-A workflow and Weather Forecast system

Directory	Contents
Inputs/Static	Mesh, static file, graph.info, and Vtable.GFS. The graph.info.part.N partitions are generated during each run.
Inputs/GFS/CICLO/FECHA	GRIB2 files downloaded from GFS.
Inputs/UNGRIBED/CICLO/FECHA	FILE:* files produced by ungrib, namelist.wps, and working links.
Ejec/CICLO/FECHA	Linked executables and permanent files, namelist, streams, init.nc, lbc, generated partitions, and links to FILE:*
Ejec/CICLO/FECHA/Logs	Download logs, ungrib logs, MPI console output, and MPAS internal logs identified by stage.
Outputs/CICLO/FECHA	history, diag, restart, and Products files.
Logs/00 y Logs/12	Symbolic links for quick access to the logs from the most recent execution of each cycle.

```

/modelos/W_frst_MPAS/
├── Scripts/
│   ├── config_sis.bash
│   ├── funciones.bash
│   ├── preparar_gfs.bash
│   ├── generar_archivos.bash
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│   └── limpiar_sistema.bash (en desarrollo)
├── Envs/
│   ├── load_mpas_gnu.bash
│   └── w_env_wrf.bash
├── FCST_OP/
│   ├── Inputs/
│   │   ├── Static/
│   │   ├── GFS/{00,12}/YYYYMMDD/
│   │   ├── UNGRIBED/{00,12}/YYYYMMDD/
│   │   └── Ejec/{00,12}/YYYYMMDD/
│   │       └── Logs/
│   ├── Outputs/{00,12}/YYYYMMDD/
│   │   └── Productos/
│   └── Logs/
├── 00 -> logs de la ejecución 00 más reciente
└── 12 -> logs de la ejecución 12 más reciente

```

MPAS-A workflow and Weather Forecast system

config_sis.bash

Variable	Value or function
DIR_MPAS	MPAS-A model installation directory
DIR_WRF	WRF model installation directory, where the WPS module must be located as a subdirectory.
DIR_STACKS	Library and tool stacks, in this case for running MPAS and WRF, as well as related tools such as qMETIS.
DIR_IMPLEMENTACION	Root directory /modelos/W_frst_MPAS.
DIR_(Varios)	Defines variables with the paths to the system internal directories

Variable	Current configuration
MODO_FECHA	Automatic: defined by the system date and time; manual: date and cycle defined by the user.
FECHA_MANUAL	E.g.: 2026-06-25, used only in manual mode.
CICLO_MANUAL	00 or 12, used only in manual mode.
HORAS_PRONOSTICO	Example: 96 hours.
INTERVALO_GFS_HORAS	1 hour.
INTERVALO_SALIDA_HORAS	1 hour.

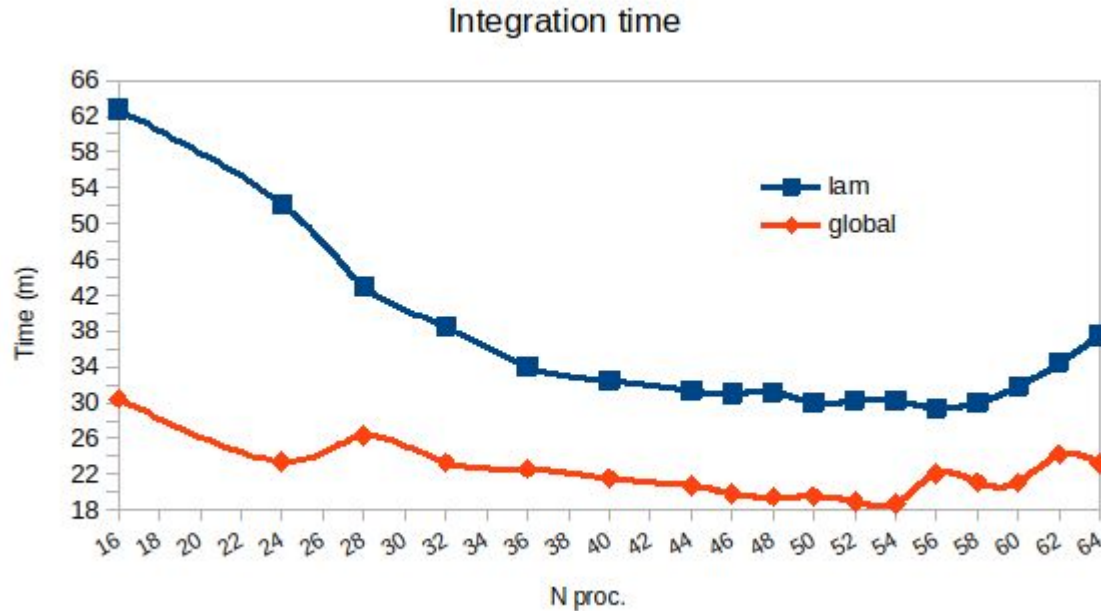
Parameter	Value
NPROC_INIT / NPROC_LBC	Number of processors allocated to generate boundary and initial conditions. E.g.: 32
NPROC_ATMOS	Number of processors allocated to run the integration. E.g.: 46
PASO_TIEMPO	30 s.
NIVELES_VERTICALES	55.
TOPE_MODELO	30 000 m.
SUITE_FISICA	convection_permitting.
RADIACION_INTERVAL	00:30:00.
INTERVALO_RESTART	1_00:00:00.

MPAS-A workflow and Weather Forecast system

Functions routine: Operational forecasting requires failure management, not only model execution.

- Input and executable validation before every stage.
- Incomplete GFS downloads stored as .part.
- Independent logs for each processing stage.
- Run directories organized by cycle/date.
- Dynamic generation of mesh partitions.
- PID-based lock prevents overlapping forecast cycles.

MPAS-A integration time vs MPI rank count



Configurations: limited-area and global
Resources: one 64-core node; 16–64 MPI ranks

Lowest measured times: approximately 54–56 ranks

At 64 ranks: longer integration time in both configurations

Lessons from forecast-system characterization

- **GFS acquisition:** connectivity failures can delay the entire forecast cycle
- **Model I/O:** excessive history output can make forecast production unsustainable
- **Storage:** hundreds of GB per cycle under the initial output configuration
- **Modular workflow:** faster identification of the failed stage
- **Reproducibility:** configuration, logs, initial conditions and lateral boundaries alongside model output

Operational adjustments and data-retention policy

- GFS downloads: retries, resume, IPv4, .part files
- History output: variables required for forecast products
- Output volume: approximately 1.3 GB/h → 32 MB/h

Retention periods

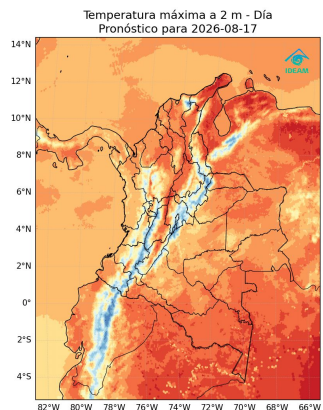
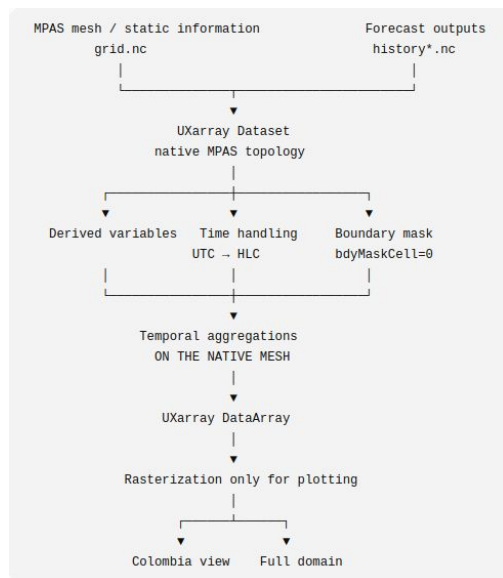
- GFS, intermediate and restart files: 2 days
- Initial and lateral boundary conditions: 5 days
- Products and logs: 45 days
- Native forecast output: at least 1 year
- Run reconstruction: preserved configuration files and logs

Products / Outputs

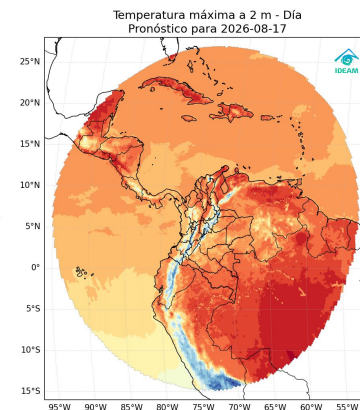
- Robust GFS download: retries, resume, IPv4 and .part files.
- Reduce history output to variables required by forecast products.
- Output reduction: $\sim 1.3 \text{ GB/h} \rightarrow \sim 32 \text{ MB/h}$.
- Differential retention:
 - GFS / intermediate / restart $\rightarrow 2 \text{ days}$
 - Initial and boundary conditions $\rightarrow 5 \text{ days}$
 - Logs and products $\rightarrow 45 \text{ days}$
 - Native forecast outputs $\rightarrow \geq 1 \text{ year}$
 - Preserve configurations and logs to support diagnosis and reproducibility.

Forecast products: Unstructured mesh challenge

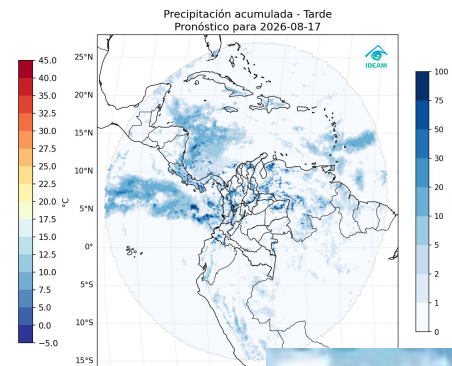
Python y UXARRAY, XARRAY, Matplotlib



IDEAM-Sub. Meteorología-GMTC | MPAS-A Res: ~23.6 Km | Elaborado:2026-08-16



IDEAM-Sub. Meteorología-GMTC | MPAS-A Res: ~23.6 Km | Elaborado:2026-08-16



IDEAM-Sub. Meteorología-GMTC | MPAS-A Res: ~23.6 Km | Elaborado:2026-08-16



IDEAM
Instituto de Hidrología, Meteorología
y Estudios Ambientales

Next steps

- Meteorological forecast verification
- Expanded HPC resources
- Colombia-focused global variable-resolution runs
- Multiple runs for ensemble forecasting



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